

Online Localized Conformal Prediction

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Abstract

Conformal prediction is a framework that provides valid uncertainty quantification for general models with exchangeable data. However, in the online learning and time-series settings, exchangeability is not satisfied. Existing online conformal methods, such as adaptive conformal inference (ACI), can achieve long-run validity, yet they remain inefficient under covariate heterogeneity because they rely on global calibration. We propose *Online Localized Conformal Prediction (OLCP)*, which combines online adaptation with covariate-dependent localization to better reflect heterogeneity. To reduce sensitivity to the localization bandwidth, we further develop *OLCP-Hedge*, which performs bandwidth selection as an online expert aggregation problem using a constrained online convex optimization framework. Importantly, we provide coverage guarantees for both algorithms and demonstrate through simulations and real-data experiments that the proposed methods attain valid long-run coverage with narrower prediction sets than existing baselines.

1 Introduction

Reliable uncertainty quantification is essential in online learning and time-series prediction, where data are often temporally dependent, nonstationary, and heterogeneous across covariate space. Our goal is to construct online prediction set $C_t(X_t) \subseteq \mathcal{Y}$ for a sequential data stream $\{(X_t, Y_t)\}_{t=1}^T$ such that the long-run coverage target

$$\frac{1}{T} \sum_{t=1}^T \mathbf{1}\{Y_t \in C_t(X_t)\} \approx 1 - \alpha \quad (1)$$

is achieved while the average set size remains as small as possible.

Conformal prediction (CP) provides distribution-free uncertainty quantification with finite-sample marginal coverage under exchangeability [28, 16, 1]. However, exchangeability is often violated in online and time-series settings due to temporal dependence, distribution shift, heteroskedasticity, and structural breaks. Existing online conformal methods such as adaptive conformal inference (ACI) [7] and dynamically tuned ACI (DtACI) [8] address this issue by adapting the nominal miscoverage level over time, and can achieve long-run validity under non-exchangeability. That being said, these methods remain *globally* calibrated: they react to temporal changes in overall uncertainty, but do not adapt set size to local covariate-dependent heterogeneity. As a result, they can be simultaneously too wide in easy regions and too narrow in difficult regions of the covariate space.

A complementary line of work addresses heterogeneity through localization. In particular, localized conformal prediction (LCP) re-weights calibration points according to covariate similarity, allowing set size to vary with local uncertainty [9]. While attractive in heterogeneous regression problems, existing localized conformal methods rely on exchangeability and are not designed for online nonstationary data streams. This leaves a natural gap: current online conformal methods handle temporal non-exchangeability but remain global, whereas current localized conformal methods adapt to heterogeneity but do not handle non-exchangeable online data.

We propose *Online Localized Conformal Prediction (OLCP)*, which combines online calibration with covariate-dependent localization. At each time t , OLCP computes a localized conformal quantile

around the current covariate X_t , and then updates the nominal miscoverage level using realized coverage feedback. Thus the set size adapts both over time and across the covariate space. A key practical issue is bandwidth selection. To reduce sensitivity to this choice, we formulate bandwidth selection as an online expert aggregation problem with prediction set size as the objective and coverage as the constraint, and develop *OLCP-Hedge*, a constrained online convex optimization procedure over a collection of OLCP experts.

Our main contributions are as follows:

- We introduce OLCP, a localized online conformal method that combines covariate-dependent calibration with feedback-driven online adaptation.
- We show that OLCP enjoys a long-run coverage guarantee under its sequential coverage-tracking update, despite replacing global conformal quantiles with localized ones.
- We formulate OLCP bandwidth selection as a constrained online convex optimization problem and propose OLCP-Hedge, which controls long-run coverage violation while competing with the best feasible bandwidth expert in terms of set size. Both coverage and set size guarantees for OLCP-Hedge are provided.
- Through simulations and real-data experiments, we show that OLCP and OLCP-Hedge achieve valid long-run coverage with narrower prediction sets than existing online conformal baselines in heterogeneous and nonstationary settings.

2 Problem setup and related work

We observe a sequential stream $\{(X_t, Y_t)\}_{t=1}^T$, where $X_t \in \mathcal{X}$ is observed before prediction and $Y_t \in \mathcal{Y}$ is revealed afterward. At each time t , using the past data together with the current covariate X_t , we construct a prediction set

$$C_t(X_t) \subseteq \mathcal{Y}$$

for Y_t . Our objective is to achieve the target long-run coverage level (1) while keeping the prediction sets as small as possible. In contrast to the classical conformal setting, we do not assume exchangeability of $\{(X_t, Y_t)\}$; the sequence may be temporally dependent, nonstationary, or heterogeneous across the covariate space.

Conformal prediction (CP). We first introduce the split conformal prediction framework [28, 1]. One uses an *independent* training sample to construct a score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, where larger values indicate that a candidate pair (x, y) is less conforming to the training data; for example, one may take $s(x, y) = |y - \hat{f}(x)|$, where $\hat{f}$ is a pretrained predictor.

Given calibration points $Z_i = (X_i, Y_i)$, $i = 1, \dots, n$, define the calibration scores $S_i := s(X_i, Y_i)$ and the empirical score distribution

$$\hat{F}_n := \frac{1}{n} \sum_{i=1}^n \delta_{S_i}, \quad \text{where } \delta_s \text{ is a point mass at } s \in \mathbb{R}.$$

For any $\tau \in [0, 1]$, let

$$Q(\tau; F) := \inf\{t \in \mathbb{R} \cup \{\infty\} : F((-\infty, t]) \geq \tau\}$$

denote the lower τ -quantile of a distribution F . Split conformal prediction calibrates the score threshold by

$$\hat{q}_{1-\alpha} := Q\left(\frac{\lceil (n+1)(1-\alpha) \rceil}{n}; \hat{F}_n\right),$$

and outputs the prediction set

$$C^{\text{CP}}(X_{n+1}) = \{y \in \mathcal{Y} : s(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}.$$

Under exchangeability of the calibration points and the test point, this construction yields finite-sample marginal coverage,

$$\mathbb{P}\{Y_{n+1} \in C^{\text{CP}}(X_{n+1})\} \geq 1 - \alpha,$$

while allowing complete flexibility in the choice of score s [16, 21].

Localized conformal prediction (LCP). A natural extension of split conformal prediction is to replace the unweighted empirical score distribution with a weighted one, assigning larger weight to calibration points that are more relevant to the test point. This idea underlies weighted conformal methods for covariate shift and related distribution shifts [13, 2]. Localized conformal prediction (LCP) specializes this weighting scheme to the covariate space, emphasizing calibration points whose features are close to the test covariate X_{n+1} [9]. Specifically, LCP introduces a localizer $H : \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$, and defines normalized local weights

$$p_{n+1,j}^H := \frac{H(X_{n+1}, X_j)}{\sum_{k=1}^{n+1} H(X_{n+1}, X_k)}, \quad j = 1, \dots, n+1,$$

together with the weighted augmented score distribution

$$\hat{F}_{n+1}^H := \sum_{j=1}^n p_{n+1,j}^H \delta_{S_j} + p_{n+1,n+1}^H \delta_{\infty}.$$

A naive localized analogue of split conformal prediction would use the weighted quantile $Q(1 - \alpha; \hat{F}_{n+1}^H)$, but this generally fails to preserve finite-sample validity. [9] shows that validity can be restored by replacing $1 - \alpha$ with a recalibrated level $\tilde{\alpha}$, leading to the prediction set

$$C^{\text{LCP}}(X_{n+1}) = \{y \in \mathcal{Y} : s(X_{n+1}, y) \leq Q(\tilde{\alpha}; \hat{F}_{n+1}^H)\}.$$

This retains finite-sample marginal coverage while allowing the prediction set to adapt to covariate-dependent heterogeneity.

Adaptive conformal inference (ACI). A different line of work addresses non-exchangeability through online calibration. Following [7], let D_t denote an estimated conformity-score distribution at time t , for example the empirical distribution of recent scores. For any $\beta \in [0, 1]$, define

$$C_t^{\text{ACI}}(\beta) = \{y \in \mathcal{Y} : s(X_t, y) \leq Q(1 - \beta; D_t)\}.$$

Adaptive conformal inference (ACI) replaces the fixed nominal level α by an online-updated parameter α_t^{ACI} . After predicting with $C_t^{\text{ACI}}(\alpha_t^{\text{ACI}})$, it observes

$$\text{err}_t^{\text{ACI}} := \mathbf{1}\{Y_t \notin C_t^{\text{ACI}}(\alpha_t^{\text{ACI}})\},$$

and updates

$$\alpha_{t+1}^{\text{ACI}} = \alpha_t^{\text{ACI}} + \gamma(\alpha - \text{err}_t^{\text{ACI}}),$$

where $\gamma > 0$ is a step size [7]. This yields a feedback-driven calibration rule that targets long-run coverage under arbitrary distribution shifts.

Other related work. A key limitation of ACI is its sensitivity to the step size γ . DtACI [8] and AgACI [32] address this by aggregating multiple ACI experts online, while recent work develops parameter-free online conformal updates based on universal portfolio algorithms [18]. Other sequential conformal methods update scores or interval sizes over time: EnbPI [29] builds prediction sets around bootstrap ensemble predictors under weak dependence conditions, and SPCI [30] forecasts future residual quantiles from past residuals. A different direction anticipates abrupt shifts using additional structure: CPTC [26] maintains state-specific score sets and aggregates them using a predicted latent state sequence, but relies on a state predictor and state-conditioned forecaster. Under exchangeability, selection and aggregation methods choose among conformal predictors while preserving finite-sample validity through recalibration [31, 17]; our setting instead treats coverage as a long-run online constraint. Finally, conformal methods beyond exchangeability use data-dependent or fixed weights, including covariate-shift conformal prediction [27] and fixed-weight approaches for non-exchangeable data [2].

3 Methods

3.1 Online Localized Conformal Prediction

We now introduce our first and main method *Online Localized Conformal Prediction* (OLCP). The starting point is that two distinct difficulties arise in sequential prediction. First, under covariate heterogeneity, uncertainty can vary substantially across the feature space, so a global conformal quantile may allocate set size inefficiently. Second, under temporal distribution shift, the appropriate calibration level itself changes over time. OLCP addresses both issues simultaneously by combining localized calibration in X with online updates of the nominal level.

Algorithm 1 Online Localized Conformal Prediction (OLCP)

Require: target miscoverage α , step size γ , bandwidth h , window length R

- 1: Initialize $\alpha_1 \in [0, 1]$
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: Compute localized weights $w_{t,i}^{(h)}(X_t)$ for $i \in \mathcal{I}_t = \{\max(1, t - R), \dots, t - 1\}$ from (2)
 - 4: Form $D_t^{(h)}(X_t) = \sum_{i \in \mathcal{I}_t} w_{t,i}^{(h)}(X_t) \delta_{S_i}$
 - 5: Output $C_t^{(h)}(X_t; \alpha_t) = \{y : s(X_t, y) \leq Q(1 - \alpha_t; D_t^{(h)}(X_t))\}$
 - 6: Observe Y_t and set $\text{err}_t = \mathbf{1}\{Y_t \notin C_t^{(h)}(X_t; \alpha_t)\}$
 - 7: Update $\alpha_{t+1} = \Pi_{[0,1]}(\alpha_t + \gamma(\alpha - \text{err}_t))$
 - 8: **end for**
-

Localized calibration distribution. Fix a bandwidth $h > 0$, a localizer

$$H_h : \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty),$$

and a rolling calibration window $\mathcal{I}_t := \{\max(1, t - R), \dots, t - 1\}$ with fixed window size R . For each $i \in \mathcal{I}_t$, $S_i = s(X_i, Y_i)$ denotes the realized conformity score at time i . For a query covariate $x \in \mathcal{X}$, define normalized local weights

$$w_{t,i}^{(h)}(x) := \frac{H_h(x, X_i)}{\sum_{j \in \mathcal{I}_t} H_h(x, X_j)}. \quad (2)$$

We then form the localized empirical distribution

$$D_t^{(h)}(x) := \sum_{i \in \mathcal{I}_t} w_{t,i}^{(h)}(x) \delta_{S_i}.$$

This is the online analogue of the weighted empirical distributions used in localized conformal prediction [9]: past scores associated with covariates close to the query point x receive more weight.

For any query covariate $x \in \mathcal{X}$ and any $\beta \in [0, 1]$, define

$$C_t^{(h)}(x; \beta) := \{y \in \mathcal{Y} : s(x, y) \leq Q(1 - \beta; D_t^{(h)}(x))\}.$$

Thus h controls the degree of localization: small h emphasizes nearby covariates, while large h recovers a more global rule [9].

OLCP update. OLCP maintains a nominal level $\alpha_t \in [0, 1]$. At time t , it outputs $C_t^{(h)}(X_t; \alpha_t)$, observes the error

$$\text{err}_t := \mathbf{1}\{Y_t \notin C_t^{(h)}(X_t; \alpha_t)\},$$

and updates

$$\alpha_{t+1} = \Pi_{[0,1]}(\alpha_t + \gamma(\alpha - \text{err}_t)), \quad (3)$$

where $\Pi_{[0,1]}$ is projection onto $[0, 1]$ and $\gamma > 0$ is a step size.

The algebraic form of (3) is similar to ACI, but the object being calibrated is different: OLCP updates the level for the localized family $\{C_t^{(h)}(X_t; \beta) : \beta \in [0, 1]\}$, not for a single global conformal rule. This distinction is what allows OLCP to adapt set size across both time and covariate space.

Pinball-loss view and coverage guarantee. The update in (3) also admits a useful optimization interpretation. Define the coverage-boundary level

$$\beta_t^{(h)} := \sup\{\beta \in [0, 1] : Y_t \in C_t^{(h)}(X_t; \beta)\},$$

namely, the boundary nominal level for which the realized response starts to be uncovered. Because $C_t^{(h)}(X_t; \beta)$ is decreasing in β ,

$$\alpha_t < \beta_t^{(h)} \Rightarrow \text{err}_t = 0, \quad \alpha_t > \beta_t^{(h)} \Rightarrow \text{err}_t = 1.$$

At $\alpha_t = \beta_t^{(h)}$, either outcome may occur. For pinball loss

$$\ell(\beta, \theta) = \alpha(\beta - \theta) - \min\{0, \beta - \theta\},$$

we therefore have

$$\text{err}_t - \alpha \in \partial_\theta \ell(\beta_t^{(h)}, \theta) \big|_{\theta=\alpha_t},$$

so (3) is exactly projected online gradient descent on the loss sequence $\{\ell(\beta_t^{(h)}, \cdot)\}_{t=1}^T$.

We introduce the following long-term coverage guarantee for OLCP with proof deferred in Appendix A.

Proposition 3.1 (Boundary-corrected coverage control for OLCP). *Define $z_t := \alpha_t + \gamma(\alpha - \text{err}_t)$, and the lower- and upper-boundary projection corrections $L_t := (-z_t)_+$, $U_t := (z_t - 1)_+$. Equivalently, $\alpha_{t+1} = z_t + L_t - U_t$. Then, for every $T \geq 1$ and fixed $\gamma > 0$,*

$$\sum_{t=1}^T (\text{err}_t - \alpha) = \frac{\alpha_1 - \alpha_{T+1}}{\gamma} + \frac{1}{\gamma} \sum_{t=1}^T (L_t - U_t).$$

In particular, if $\sum_{t=1}^T L_t = o(T)$, then $\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \text{err}_t \leq \alpha$.

If additionally $\sum_{t=1}^T (L_t + U_t) = o(T)$, then $\frac{1}{T} \sum_{t=1}^T \text{err}_t \rightarrow \alpha$.

Remark on boundary terms. Proposition 3.1 shows exactly what projection changes relative to the unprojected ACI. Unlike ACI, which can use infinite or empty prediction sets to obtain exact telescoping, OLCP keeps prediction sets finite and must account for L_t and U_t . The lower correction L_t measures unresolved undercoverage pressure: it is positive only when the update wants to move α_t below 0, i.e. when OLCP misses while already using a very wide finite set. The upper correction U_t is the analogous overcoverage pressure near $\alpha_t = 1$. Thus projected OLCP recovers ACI-style long-run calibration whenever these normalized boundary corrections are negligible. Appendix B gives sufficient conditions under which these terms vanish and explains how to diagnose them empirically.

3.2 OLCP-Hedge: constrained aggregation across localization bandwidths

OLCP requires choosing a localization bandwidth h . This bandwidth controls a familiar bias–variance trade-off: small h adapts strongly to local heterogeneity but can be noisy, whereas large h is more stable but may wash out covariate-dependent uncertainty [9]. Since the best localization scale is generally unknown and may vary over time, this motivates aggregating a collection of OLCP experts rather than committing to a single bandwidth in advance.

This is related to the classical prediction-with-expert-advice problem, where the learner combines expert predictions and competes with the best expert in hindsight [19]. However, our objective is not a single scalar loss. If set size is used as the loss, ordinary Hedge may favor narrow experts that undercover; if miscoverage is used as the loss, the resulting prediction sets may be valid but inefficient. Thus bandwidth selection is naturally a constrained online problem: minimize set size while controlling long-run miscoverage.

OLCP expert pool. Fix K OLCP procedures, for example, with different bandwidths $\{h_i : i = 1, \dots, K\}$. At time t , expert i outputs a prediction set $C_{t,i}(X_t)$, for example $C_{t,i}(X_t) := C_t^{(h_i)}(X_t; \alpha_t)^1$, and receives feedback of

$$\text{size } \omega_{t,i} := \text{size}(C_{t,i}(X_t)) \quad \text{and} \quad \text{miscoverage } \text{err}_{t,i} := \mathbf{1}\{Y_t \notin C_{t,i}(X_t)\},$$

where $\text{size}(\cdot)$ could be interval width, or cardinality depending on the tasks.

Let $\omega_t = (\omega_{t,1}, \dots, \omega_{t,K})$ and $e_t = (\text{err}_{t,1}, \dots, \text{err}_{t,K})$. We maintain a distribution $p_t \in \Delta_K$ over experts, sample $I_t \sim p_t$, and output $C_{t,I_t}(X_t)$,

Size objective and miscoverage constraint. For $p \in \Delta_K$, define

$$f_t(p) := \langle \omega_t, p \rangle, \quad g_t(p) := \langle e_t, p \rangle - \alpha.$$

Then

$$\mathbb{E}_{I_t \sim p_t} [\omega_{t,I_t}] = f_t(p_t), \quad \mathbb{E}_{I_t \sim p_t} [\text{err}_{t,I_t}] = \langle e_t, p_t \rangle = g_t(p_t) + \alpha.$$

Thus $g_t(p_t)$ is the expected excess miscoverage of the aggregate. Equivalently, the meta-objective is

$$\min_{p_t \in \Delta_K} \sum_{t=1}^T f_t(p_t) \quad \text{while keeping} \quad \sum_{t=1}^T (g_t(p_t))_+ \text{ sublinear.}$$

This is an instance of *constrained online convex optimization* (COCO): the learner chooses an action before observing a convex loss and constraint, and aims to control both regret and cumulative constraint violation. COCO has a substantial recent literature on time-varying constraints and long-term feasibility; see, e.g., [10, 24] for recent overviews and comparisons.

¹Our algorithm can also be applied to aggregate OLCP procedures with different step size γ , see more on Section 5.

Algorithm 2 OLCP-Hedge

Require: target miscoverage α , OLCP experts $\{\{C_{t,i}\}_{i=1}^K\}_{t=1}^T$, parameters V, κ, λ

- 1: Initialize $\mathcal{Q}(0) = 0$ and AdaHedge on Δ_K
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: AdaHedge outputs $p_t \in \Delta_K$
- 4: Sample $I_t \sim p_t$ and output $C_{t,I_t}(X_t)$
- 5: Observe Y_t ; compute the size $\omega_{t,i}$ and $\text{err}_{t,i} = \mathbf{1}\{Y_t \notin C_{t,i}(X_t)\}$ for all i
- 6: Define $f_t(p) = \langle \omega_t, p \rangle$, $g_t(p) = \langle e_t, p \rangle - \alpha$
- 7: Update $\mathcal{Q}(t) = \mathcal{Q}(t-1) + \kappa(g_t(p_t))_+$
- 8: Form $\hat{f}_t(p) = V\kappa f_t(p) + \Phi'(\mathcal{Q}(t))\kappa(g_t(p))_+$
- 9: Choose $\xi_t \in \partial \hat{f}_t(p_t)$ and feed the linearized loss $\ell_t(p) = \langle \xi_t, p \rangle$ to AdaHedge
- 10: **end for**

We adapt the algorithm proposed in [24] because it gives state-of-the-art simultaneous guarantees, $O(\sqrt{T})$ regret and $\tilde{O}(\sqrt{T})$ cumulative constraint violation, without Slater-type assumptions or per-round constrained optimization. Their reduction combines a Lyapunov-style virtual queue with a black-box adaptive OCO subroutine.

The original paper focuses on the Euclidean space; here we extend their algorithm to the probability simplex of OLCP experts, and prove the corresponding size-regret and excess-miscoverage bounds in the probability simplex geometry.

Assumptions. We use the following assumptions, specialized to our expert aggregation problem.

- **Assumption A (bounded sizes).** There exists $G > 0$ such that for all t ,

$$\|\omega_t\|_\infty \leq G, \quad \|e_t\|_\infty \leq G.$$

Since $e_{t,i} \in \{0, 1\}$, this only requires a uniform bound on set sizes. This assumption is mild after size normalization or when the response range is bounded.

- **Assumption B (uniformly feasible comparator).** There exists $u^* \in \Delta_K$ such that

$$g_t(u^*) = \langle e_t, u^* \rangle - \alpha \leq 0, \quad \forall t.$$

This assumption is a standard but strong feasibility condition in COCO: one compares against a fixed feasible action. In our setting, it can be enforced by including a conservative expert, although this may increase the comparator size. If exact feasibility is unavailable, the same viewpoint suggests a relaxed fallback using online constraint-satisfaction ideas from [24]: instead of requiring a uniformly feasible comparator, one can compare to an S -feasible or P_T -constrained benchmark and aim for sublinear excess miscoverage relative to that weaker benchmark. We discuss diagnostics for this assumption in Appendix C.

Surrogate loss. Following [24], OLCP-Hedge maintains a virtual queue for excess miscoverage and feeds AdaHedge¹[5] a surrogate loss that combines set size with queue-weighted constraint violation. Let $\mathcal{Q}(0) = 0$ and update

$$\mathcal{Q}(t) = \mathcal{Q}(t-1) + \kappa(g_t(p_t))_+, \quad \text{where } \kappa > 0 \text{ is a constant.}$$

With $\Phi(q) = e^{\lambda q} - 1$ and $\lambda > 0$, define

$$\hat{f}_t(p) = V\kappa f_t(p) + \Phi'(\mathcal{Q}(t))\kappa(g_t(p))_+, \quad \text{where } V > 0 \text{ is a constant.}$$

The first term penalizes prediction set size, while the second term increasingly penalizes excess miscoverage as the queue grows. The full algorithm is detailed in Algorithm 2 with more details of AdaHedge in Appendix D.

Guarantee. The next theorem states that OLCP-Hedge competes with the best feasible expert mixture in expected size while keeping cumulative expected excess miscoverage sublinear. The proof is deferred to Appendix E.

¹AdaHedge is only one possible subroutine; see the modularity discussion below in 3.2.

Theorem 3.2 (Size regret and excess-miscoverage control). *Assume Assumptions A–B. Run Algorithm 2 with*

$$V = 1, \quad C_{\text{AH}} := 2\sqrt{4 + \ln K}, \quad \kappa := \frac{1}{\sqrt{2} C_{\text{AH}} G}, \quad \lambda := \frac{1}{2\sqrt{T}}.$$

Then, for any feasible comparator $u^ \in \Delta_K$, we have the following expected set-size regret bound:*

$$\sum_{t=1}^T \left(\mathbb{E}_{I_t \sim p_t} [\omega_{t, I_t}] - \langle \omega_t, u^* \rangle \right) \leq 4G\sqrt{2(4 + \ln K)T} = O\left(G\sqrt{T(1 + \ln K)}\right),$$

and the following cumulative excess-miscoverage bound:

$$\sum_{t=1}^T \left(\mathbb{E}_{I_t \sim p_t} [\text{err}_{t, I_t}] - \alpha \right)_+ \leq 4G\sqrt{2(4 + \ln K)T} \ln\left(2 + \left(2 + \frac{\sqrt{2}}{2}\right)T\right) = \tilde{O}\left(G\sqrt{T(1 + \ln K)}\right).$$

Remark on the modularity of the subroutine. The OLCP-Hedge analysis is modular in the OCO subroutine, following the black-box philosophy of [24]. The proof only uses that, when the subroutine is fed the linearized surrogate losses $\ell_t(p) = \langle \xi_t, p \rangle$, it satisfies a data-dependent regret bound of the form

$$\sum_{t=1}^T \langle \xi_t, p_t - u \rangle \leq C \sqrt{\sum_{t=1}^T \|\xi_t\|_\infty^2}, \quad \forall u \in \Delta_K.$$

AdaHedge is one parameter-free expert algorithm with such a worst-case/adaptive guarantee [5, 19]. More broadly, related adaptive expert algorithms providing comparable data-dependent regret guarantees could be used in the same reduction after replacing C_{AH} by the corresponding constant or regret complexity [6, 14, 20]. Thus the theorem is not tied to AdaHedge itself.

4 Experiments

We evaluate OLCP and OLCP-Hedge on both synthetic experiments and real sequential prediction tasks. We first introduce our experimental setup, then present simulation and real-data results. Code is available online.

4.1 Experimental setup

We compare seven methods (see Section 2): CP [16], LCP [9], ACI [7], DtACI [8], SPCI [30], and proposed OLCP/OLCP-Hedge. All methods use the same base predictor, conformity scores, and rolling calibration window. For localized methods (LCP/OLCP/OLCP-Hedge), we use an exponential kernel

$$H_h(x, x') = \exp\left(-\frac{\|\tilde{x} - \tilde{x}'\|_2}{h}\right),$$

where covariates are standardized within the current calibration window before computing distances. The base bandwidth h_0 is chosen by a Silverman-style rule, and OLCP-Hedge aggregates the grid

$$h \in \{0.5, 0.75, 1, 1.25, 1.5\}h_0.$$

All calibration is online: at time t , each method uses only past conformity scores in the rolling window. Full implementation details, including bandwidth formulas, adaptive step sizes, and SPCI settings, are given in Appendix F.1.

All methods are evaluated using empirical coverage and average set size, for prediction set (interval) $C_t(X_t) = [L_t(X_t), U_t(X_t)]$ produced by any methods above, we measure

$$\widehat{\text{cov}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{Y_t \in C_t(X_t)\}, \quad \widehat{\text{size}} = \frac{1}{T} \sum_{t=1}^T (U_t(X_t) - L_t(X_t)).$$

4.2 Simulations

We first use controlled synthetic experiments to isolate two failure modes of existing online conformal methods: covariate-dependent heterogeneity and abrupt temporal distribution shift. We simulate a univariate time series $\{Y_t\}_{t=0}^T$ and form a one-step-ahead prediction task with $X_t = Y_{t-1}$. A fixed linear predictor $\hat{f}$ is trained on the first 500 observations, and all methods use the absolute residual score $S_t = |Y_t - \hat{f}(X_t)|$. We use $T = 1,500$, rolling calibration window $R = 200$, target miscoverage $\alpha = 0.1$, and 100 Monte Carlo repetitions. All the results are reported on the test set.

Let $\varepsilon_t \sim N(0, 1)$. We consider three scenarios:

Table 1: Simulation results over 100 repetitions. Each entry reports mean (standard deviation). Boldface marks the smallest average size among methods whose coverage attains the 0.90 target.

Method	A: Stationary		B: Heterogeneous		C: Change point	
	Coverage	Size	Coverage	Size	Coverage	Size
CP	0.900 (0.005)	3.301 (0.086)	0.900 (0.005)	3.611 (0.142)	0.869 (0.009)	7.363 (0.468)
LCP	0.892 (0.005)	3.271 (0.085)	0.894 (0.006)	3.448 (0.106)	0.879 (0.008)	6.083 (0.289)
ACI	0.900 (0.003)	3.341 (0.095)	0.900 (0.003)	3.738 (0.192)	0.900 (0.004)	7.908 (0.477)
DtACI	0.900 (0.003)	3.326 (0.094)	0.900 (0.004)	3.692 (0.171)	0.898 (0.004)	7.840 (0.466)
SPCI	0.812 (0.010)	2.850 (0.070)	0.819 (0.009)	3.056 (0.109)	0.773 (0.014)	3.371 (0.116)
OLCP	0.900 (0.003)	3.382 (0.100)	0.901 (0.003)	3.572 (0.129)	0.901 (0.003)	6.349 (0.278)
OLCP-Hedge	0.900 (0.003)	3.376 (0.097)	0.900 (0.003)	3.564 (0.128)	0.900 (0.004)	6.348 (0.278)

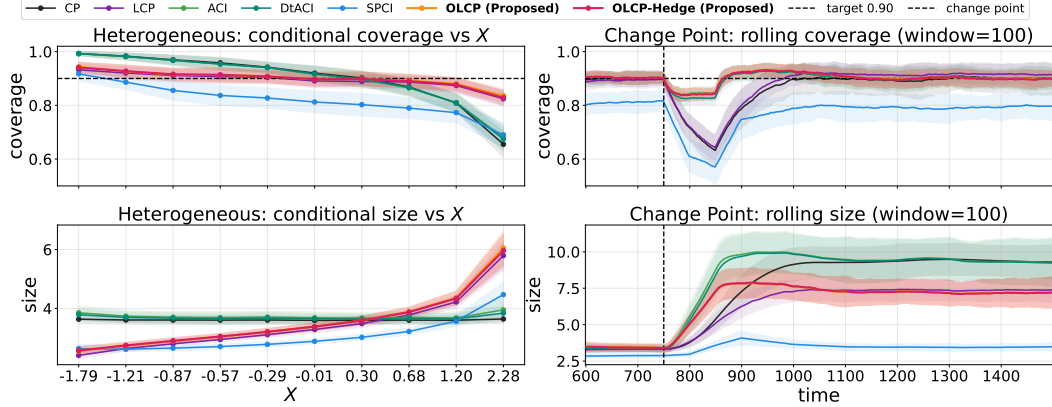


Figure 1: **Diagnostics for simulation.** Left panels show Scenario B conditional coverage and average size across X_t . Right panels show Scenario C rolling coverage and rolling average size with window size 100; the vertical dashed line marks the change point. Shaded bands show mean $\pm$ one standard deviation across repetitions. OLCP (DtACI) curves are partially hidden behind OLCP-Hedge (ACI).

- **A: Stationary:** $Y_t = 0.5Y_{t-1} + \varepsilon_t$.
- **B: Heterogeneous:** $Y_t = 0.5Y_{t-1} + \sigma_t \varepsilon_t$, where $\sigma_t = \min\{\exp(0.25Y_{t-1}), 10\}$.
- **C: Change point:** $Y_t = \phi_t Y_{t-1} + \varepsilon_t$, where $\phi_t = \begin{cases} 0.8, & t \leq T/2, \\ -0.8, & t > T/2. \end{cases}$

Scenario A is a sanity check, Scenario B tests adaptation to covariate-dependent noise, and Scenario C tests adaptation to abrupt temporal shift.

Results. Table 1 summarizes marginal coverage and average prediction set size, with running time of each method reported in Appendix F.2. Scenario A is a sanity check: in the stationary homoskedastic case, there is little structure for either localization or adaptation to exploit, and the valid methods have similar sizes.

Scenario B isolates covariate-dependent heterogeneity. Global methods (CP, ACI, DtACI) achieve reasonable marginal coverage, but Figure 1 shows that their prediction-set sizes are nearly constant in X_t , causing undercoverage in high-noise regions. LCP localizes the residual quantile but mildly undercovers due to its fixed nominal level. OLCP and OLCP-Hedge combine localization with online calibration, expanding sets where noise is high while maintaining near-nominal marginal coverage; OLCP-Hedge gives the smallest size among near-valid methods.

Scenario C isolates abrupt temporal shift. CP and LCP recover slowly after the change point and undercover, while ACI and DtACI restore coverage mainly through global size inflation. OLCP and OLCP-Hedge recover to the target coverage level with substantially smaller sets, showing that localization improves efficiency under heterogeneous uncertainty and online calibration restores validity under temporal shift.

4.3 Real-data experiments

We evaluate the proposed methods on three real time-series datasets; additional preprocessing details, model hyperparameters, data splits, and diagnostic plots are deferred to Appendix F.3.

Table 2: Real-data results. Each entry reports mean (block-bootstrap standard error). Coverage is empirical marginal coverage; Size is average interval width. For ETF volatility, Size is reported as percentage points of absolute daily log return. ELEC2 and ILINet sizes are in their normalized response scales.

Method	ELEC2		ILINet		ETF volatility	
	Coverage	Size	Coverage	Size	Coverage	Size (%)
CP	0.890 (0.007)	0.3513 (0.008)	0.897 (0.034)	0.9366 (0.134)	0.905 (0.005)	2.6890 (0.062)
LCP	0.896 (0.006)	0.3312 (0.007)	0.908 (0.030)	0.7810 (0.093)	0.896 (0.005)	2.4269 (0.053)
ACI	0.900 (0.006)	0.3675 (0.009)	0.900 (0.033)	0.9004 (0.142)	0.901 (0.004)	2.6741 (0.088)
DtACI	0.898 (0.006)	0.3660 (0.009)	0.900 (0.033)	0.8886 (0.141)	0.900 (0.004)	2.6178 (0.082)
SPCI	0.777 (0.008)	0.1780 (0.002)	0.794 (0.043)	0.5110 (0.075)	0.842 (0.006)	2.2495 (0.056)
OLCP	0.900 (0.005)	0.3377 (0.008)	0.904 (0.027)	0.7917 (0.110)	0.900 (0.004)	2.5587 (0.083)
OLCP-Hedge	0.899 (0.005)	0.3351 (0.008)	0.900 (0.027)	0.7478 (0.105)	0.900 (0.004)	2.5516 (0.079)

- **ELEC2.** ELEC2 contains electricity market prices, demands, and transfers from New South Wales and Victoria [11]. We predict transfer using the two states’ prices and demands as covariates. After removing the initial constant-response segment, the dataset has 27,552 rows; the base predictor is a gradient-boosted regression tree.
- **ILINet.** ILINet is a weekly CDC influenza-like illness surveillance dataset with 1,305 observations [4, 12]. We predict the state-population-weighted weekly patient percentage using a lag window of 26 past responses as covariates, with a temporal convolutional network (TCN) as the base predictor [15].
- **ETF volatility.** We forecast absolute daily log returns for five ETFs in recent 18 years (4,742 trading days): SPY, QQQ, IWM, EEM, and TLT [25]. Covariates are a lag window of 30 past observations and the lagged VIX index [3]; the base predictor is a TCN. Widths are reported in percentage points of absolute log return.

Table 2 summarizes the real-data results with running time in Appendix F.2. Standard errors are computed using a block bootstrap over time, with block lengths 48/26/20 for ELEC2/ILINet/ETF volatility, respectively and 1000 bootstrap replicates. Across datasets, fixed LCP often yields shorter intervals but can undercover, while ACI and DtACI restore nominal coverage at the cost of wider intervals. OLCP and OLCP-Hedge provide the strongest overall coverage–efficiency tradeoff: OLCP-Hedge is consistently near-nominal and is the most efficient among the near-valid methods.

The rolling diagnostics in Appendix F.3 further clarify this pattern. Figures 2–4 show that ACI and DtACI tend to maintain coverage by raising interval sizes globally, whereas OLCP and OLCP-Hedge track the target with lower rolling size over much of the test period. The ETF volatility regime analysis in Table 4 gives a more targeted view: in low-volatility periods, all near-valid methods over-cover, but OLCP and OLCP-Hedge achieve the smallest sizes among them. In high-volatility periods, OLCP attains the best trade-off with coverage 0.890 with size 4.004, improving on both ACI (0.890, 4.349) and DtACI (0.889, 4.184). These diagnostics reinforce the central message: localization improves efficiency by adapting set size to the covariate regime, while online calibration maintains long-run coverage under temporal shift.

5 Conclusion and future work

Our results show that localization improves efficiency under heterogeneity, while online update restores coverage under temporal shift. However, several limitations remain. Like other localized conformal methods [9], OLCP also depends on informative covariates and a suitable distance metric. OLCP-Hedge assumes a uniformly feasible expert mixture, which may require conservative experts and can be restrictive in unbounded regression; it also uses full-information feedback and could add to computational burden. Future work could extend constrained aggregation to step sizes, localizers, distance metrics, calibration windows, and base predictors, and study learned localization, bandit-feedback variants, and stronger local coverage guarantees.

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A Proof of Proposition 3.1

Proof. By definition,

$$z_t = \alpha_t + \gamma(\alpha - \text{err}_t), \quad L_t = (-z_t)_+, \quad U_t = (z_t - 1)_+.$$

Since projection onto $[0, 1]$ satisfies

$$\Pi_{[0,1]}(z_t) = z_t + L_t - U_t,$$

the projected update can be written as

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t) + L_t - U_t.$$

Summing this identity over $t = 1, \dots, T$ gives

$$\alpha_{T+1} - \alpha_1 = \gamma \sum_{t=1}^T (\alpha - \text{err}_t) + \sum_{t=1}^T (L_t - U_t).$$

Rearranging yields

$$\sum_{t=1}^T (\text{err}_t - \alpha) = \frac{\alpha_1 - \alpha_{T+1}}{\gamma} + \frac{1}{\gamma} \sum_{t=1}^T (L_t - U_t),$$

which proves the claimed identity.

Since $\alpha_{T+1} \in [0, 1]$, $L_t \geq 0$, and $U_t \geq 0$, we have

$$\sum_{t=1}^T (\text{err}_t - \alpha) \leq \frac{\alpha_1}{\gamma} + \frac{1}{\gamma} \sum_{t=1}^T L_t.$$

Dividing by T gives

$$\frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \leq \frac{\alpha_1}{T\gamma} + \frac{1}{T\gamma} \sum_{t=1}^T L_t.$$

If $\gamma > 0$ is fixed and $\sum_{t=1}^T L_t = o(T)$, the right-hand side converges to zero. Hence

$$\limsup_{T \rightarrow \infty} \left(\frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right) \leq 0.$$

For the lower deviation, the same identity gives

$$\sum_{t=1}^T (\alpha - \text{err}_t) = \frac{\alpha_{T+1} - \alpha_1}{\gamma} + \frac{1}{\gamma} \sum_{t=1}^T (U_t - L_t).$$

Using $\alpha_{T+1} \leq 1$, $L_t \geq 0$, and $U_t \geq 0$, we obtain

$$\sum_{t=1}^T (\alpha - \text{err}_t) \leq \frac{1 - \alpha_1}{\gamma} + \frac{1}{\gamma} \sum_{t=1}^T U_t.$$

Dividing by T ,

$$\alpha - \frac{1}{T} \sum_{t=1}^T \text{err}_t \leq \frac{1 - \alpha_1}{T\gamma} + \frac{1}{T\gamma} \sum_{t=1}^T U_t.$$

If $\sum_{t=1}^T (L_t + U_t) = o(T)$, then in particular $\sum_{t=1}^T L_t = o(T)$ and $\sum_{t=1}^T U_t = o(T)$. Therefore both the upper and lower deviations vanish, and hence

$$\frac{1}{T} \sum_{t=1}^T \text{err}_t \rightarrow \alpha.$$

□

B More discussion on boundary terms

This section expands on Proposition 3.1. The proposition is an exact pathwise identity for the projected update

$$\alpha_{t+1} = \Pi_{[0,1]}(\alpha_t + \gamma(\alpha - \text{err}_t)).$$

The correction terms

$$L_t = (-z_t)_+, \quad U_t = (z_t - 1)_+, \quad z_t = \alpha_t + \gamma(\alpha - \text{err}_t),$$

measure how much the unconstrained update would have moved outside the interval $[0, 1]$. Thus L_t is the amount of lower-boundary clipping, and U_t is the amount of upper-boundary clipping.

Interpretation. The term $L_t > 0$ can occur only when $\text{err}_t = 1$ and

$$\alpha_t < \gamma(1 - \alpha).$$

In words, OLCP misses while α_t is already near 0. Since smaller α_t corresponds to wider prediction sets, L_t records the amount by which the update would like to widen the set beyond the widest finite set allowed by the projected implementation. Similarly, $U_t > 0$ can occur only when $\text{err}_t = 0$ and

$$\alpha_t > 1 - \gamma\alpha.$$

This records the amount by which the update would like to shrink the set beyond the smallest finite set allowed by the projection.

This is the key difference from the basic ACI telescoping guarantee. The original unprojected ACI update allows α_t to leave $[0, 1]$, which can produce full/infinite or empty prediction sets at the boundary. This makes the recursion telescope exactly without correction terms. Projected OLCP instead keeps all prediction sets finite, so the price is the appearance of L_t and U_t .

A sufficient condition. We next give a simple sufficient condition under which the boundary corrections are negligible. For clarity, allow the step size to depend on the horizon and write $\gamma = \gamma_T$. Define

$$\eta_T^- := \gamma_T(1 - \alpha), \quad \eta_T^+ := \gamma_T\alpha.$$

Since the prediction sets are decreasing in the nominal level, we have the deterministic inclusions

$$\{L_t > 0\} \subseteq \{Y_t \notin C_t^{(h)}(X_t; \eta_T^-)\},$$

and

$$\{U_t > 0\} \subseteq \{Y_t \in C_t^{(h)}(X_t; 1 - \eta_T^+)\}.$$

Moreover,

$$0 \leq L_t \leq \gamma_T(1 - \alpha) \mathbf{1}\{Y_t \notin C_t^{(h)}(X_t; \eta_T^-)\},$$

and

$$0 \leq U_t \leq \gamma_T\alpha \mathbf{1}\{Y_t \in C_t^{(h)}(X_t; 1 - \eta_T^+)\}.$$

Therefore, a sufficient condition for

$$\sum_{t=1}^T L_t = o(T\gamma_T) \quad \text{and} \quad \sum_{t=1}^T U_t = o(T\gamma_T)$$

in expectation is

$$\frac{1}{T} \sum_{t=1}^T \mathbb{P}\{Y_t \notin C_t^{(h)}(X_t; \eta_T^-) \mid \mathcal{F}_{t-1}\} = o(1),$$

and

$$\frac{1}{T} \sum_{t=1}^T \mathbb{P}\{Y_t \in C_t^{(h)}(X_t; 1 - \eta_T^+) \mid \mathcal{F}_{t-1}\} = o(1).$$

These conditions say that boundary failures are rare: the near-widest finite set should not miss too often, and the near-smallest finite set should not cover too often.

Empirical diagnostic. The boundary terms are directly observable during the run. In practice, one can report

$$\frac{1}{T\gamma} \sum_{t=1}^T L_t \quad \text{and} \quad \frac{1}{T\gamma} \sum_{t=1}^T U_t$$

as diagnostics. Small values indicate that projection is not materially affecting the long-run coverage behavior; large values indicate that the algorithm is frequently hitting the finite-set boundary, so the target coverage may not be attainable without wider or narrower endpoint sets.

C Feasibility diagnostics for assumption B

Assumption B requires a fixed mixture $u^* \in \Delta_K$ satisfying $\langle e_t, u^* \rangle \leq \alpha$ on every round. Since OLCF-Hedge uses full-information feedback, this condition can be checked post hoc by the linear feasibility problem

$$\text{find } u \in \Delta_K \quad \text{such that} \quad \langle e_t, u \rangle \leq \alpha, \quad t = 1, \dots, T.$$

Equivalently, one may solve the relaxed linear program

$$\min_{u \in \Delta_K, \rho \geq 0} \rho \quad \text{subject to} \quad \langle e_t, u \rangle - \alpha \leq \rho, \quad t = 1, \dots, T.$$

If the optimum is $\rho = 0$, Assumption B holds for the realized expert pool; otherwise ρ quantifies the worst-round infeasibility of the best fixed mixture. When it's large, the uniform-feasibility comparator required by Theorem 3.2 is inappropriate, and a relaxed online constraint-satisfaction benchmark such as S -feasibility or P_T -constrained feasibility is more suitable.

D AdaHedge subroutine

We use AdaHedge as the black-box OCO subroutine over the probability simplex Δ_K . Let

$$C_{\text{AH}} := 2\sqrt{4 + \ln K}.$$

Algorithm 3 AdaHedge on Δ_K (FTRL form; [19])

Require: parameter $\alpha_{\text{AH}} > 0$

1: Initialize

$$\lambda_1 \leftarrow 0, \quad p_1 \leftarrow (1/K, \dots, 1/K) \in \Delta_K, \quad \theta_1 \leftarrow 0 \in \mathbb{R}^K.$$

2: **for** $t = 1, 2, \dots, T$ **do**

3: Output $p_t \in \Delta_K$.

4: Observe a linear loss vector $\xi_t \in \mathbb{R}^K$ and incur loss $\langle \xi_t, p_t \rangle$.

5: Update $\theta_{t+1} \leftarrow \theta_t - \xi_t$.

6: **if** $t = 1$ **then**

7: $\delta_t \leftarrow \langle \xi_1, p_1 \rangle - \min_{j \in [K]} \xi_{1,j}$.

8: **else**

9:

$$\delta_t \leftarrow \lambda_t \ln \left(\sum_{j=1}^K p_{t,j} \exp(-\xi_{t,j}/\lambda_t) \right) + \langle \xi_t, p_t \rangle.$$

10: **end if**

11: Update $\lambda_{t+1} \leftarrow \lambda_t + \delta_t/\alpha_{\text{AH}}^2$.

12: Update

$$p_{t+1,j} \propto \exp(\theta_{t+1,j}/\lambda_{t+1}), \quad j = 1, \dots, K.$$

13: **end for**

Theorem D.1 (AdaHedge bound on Δ_K). *Run Algorithm 3 with*

$$\alpha_{\text{AH}} = \sqrt{\ln K}.$$

Then for any sequence $\{\xi_t\}_{t=1}^T \subset \mathbb{R}^K$, the iterates $p_t \in \Delta_K$ satisfy, for all $u \in \Delta_K$,

$$\sum_{t=1}^T \langle \xi_t, p_t - u \rangle \leq C_{\text{AH}} \sqrt{\sum_{t=1}^T \|\xi_t\|_\infty^2}.$$

Proof. This is the parameter-free AdaHedge guarantee obtained by optimizing the standard FTRL/AdaHedge bound; see [19, Section 7]. We use the ℓ_∞ -norm form because the decision set is the simplex and the losses are linear in p . $\square$

E Proof of Theorem 3.2

Proof. We prove a slightly stronger prefix version: the stated bounds hold for every $m \leq T$. The proof follows the Sinha–Vaze Lyapunov reduction for constrained online convex optimization, specialized to the simplex Δ_K and to the OLCP expert set size and miscoverage constraint.

Recall that

$$f_t(p) = \langle \omega_t, p \rangle, \quad g_t(p) = \langle e_t, p \rangle - \alpha,$$

where $\omega_t = (\omega_{t,1}, \dots, \omega_{t,K})$ is the vector of expert sizes and $e_t = (\text{err}_{t,1}, \dots, \text{err}_{t,K})$ is the vector of expert miscoverage indicators. We use the preprocessed size loss and excess-miscoverage constraint

$$\tilde{f}_t(p) := \kappa f_t(p), \quad \tilde{g}_t(p) := \kappa(g_t(p))_+.$$

The virtual queue is initialized at $\mathcal{Q}(0) = 0$ and updated as

$$\mathcal{Q}(t) := \mathcal{Q}(t-1) + \tilde{g}_t(p_t).$$

The potential is

$$\Phi(q) = e^{\lambda q} - 1, \quad \Phi'(q) = \lambda e^{\lambda q},$$

and the surrogate loss is

$$\hat{f}_t(p) := V \tilde{f}_t(p) + \Phi'(\mathcal{Q}(t)) \tilde{g}_t(p).$$

In Theorem 3.2, we set $V = 1$.

Step 1: Drift inequality and decomposition.

Fix any prefix length $m \leq T$. By convexity of Φ and the queue recursion

$$\mathcal{Q}(t) = \mathcal{Q}(t-1) + \tilde{g}_t(p_t),$$

we have

$$\Phi(\mathcal{Q}(t)) - \Phi(\mathcal{Q}(t-1)) \leq \Phi'(\mathcal{Q}(t)) \tilde{g}_t(p_t).$$

Summing from $t = 1$ to m and using $\Phi(\mathcal{Q}(0)) = 0$, we get

$$\Phi(\mathcal{Q}(m)) \leq \sum_{t=1}^m \Phi'(\mathcal{Q}(t)) \tilde{g}_t(p_t).$$

Let $u^* \in \Delta_K$ be any feasible comparator from Assumption B. Then

$$g_t(u^*) \leq 0 \quad \Rightarrow \quad (g_t(u^*))_+ = 0 \quad \Rightarrow \quad \tilde{g}_t(u^*) = 0.$$

Since $V = 1$,

$$\sum_{t=1}^m (\hat{f}_t(p_t) - \hat{f}_t(u^*)) = \sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) + \sum_{t=1}^m \Phi'(\mathcal{Q}(t)) \tilde{g}_t(p_t).$$

Therefore,

$$\Phi(\mathcal{Q}(m)) + \sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \leq \sum_{t=1}^m (\hat{f}_t(p_t) - \hat{f}_t(u^*)). \quad (4)$$

Step 2: Apply AdaHedge to the linearized surrogate losses.

For each t , choose any

$$\xi_t \in \partial \hat{f}_t(p_t).$$

By convexity,

$$\hat{f}_t(p_t) - \hat{f}_t(u^*) \leq \langle \xi_t, p_t - u^* \rangle.$$

Thus, by Theorem D.1 applied to the prefix $1, \dots, m$,

$$\sum_{t=1}^m (\hat{f}_t(p_t) - \hat{f}_t(u^*)) \leq C_{\text{AH}} \sqrt{\sum_{t=1}^m \|\xi_t\|_\infty^2}. \quad (5)$$

Step 3: Bound the surrogate gradients.

Since

$$\hat{f}_t(p) = \tilde{f}_t(p) + \Phi'(\mathcal{Q}(t))\tilde{g}_t(p),$$

and

$$\tilde{f}_t(p) = \kappa f_t(p), \quad \tilde{g}_t(p) = \kappa(g_t(p))_+,$$

Assumption A implies that we may choose $\xi_t \in \partial \hat{f}_t(p_t)$ such that

$$\|\xi_t\|_\infty \leq \kappa G(1 + \Phi'(\mathcal{Q}(t))).$$

Hence, using $(a+b)^2 \leq 2(a^2 + b^2)$,

$$\sum_{t=1}^m \|\xi_t\|_\infty^2 \leq 2\kappa^2 G^2 \left(m + \sum_{t=1}^m (\Phi'(\mathcal{Q}(t)))^2 \right).$$

Because $\mathcal{Q}(t)$ is nondecreasing and Φ' is nondecreasing,

$$\Phi'(\mathcal{Q}(t)) \leq \Phi'(\mathcal{Q}(m)), \quad t \leq m.$$

Therefore,

$$\sum_{t=1}^m (\Phi'(\mathcal{Q}(t)))^2 \leq m(\Phi'(\mathcal{Q}(m)))^2.$$

Plugging this into (5) gives

$$\sum_{t=1}^m (\hat{f}_t(p_t) - \hat{f}_t(u^*)) \leq \sqrt{2} C_{\text{AH}} \kappa G \sqrt{m} (1 + \Phi'(\mathcal{Q}(m))). \quad (6)$$

Step 4: Size-regret bound.

Combining (4) and (6), and using

$$\Phi(q) = e^{\lambda q} - 1, \quad \Phi'(q) = \lambda e^{\lambda q},$$

yields

$$e^{\lambda \mathcal{Q}(m)} - 1 + \sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \leq \sqrt{2} C_{\text{AH}} \kappa G \sqrt{m} (1 + \lambda e^{\lambda \mathcal{Q}(m)}).$$

With

$$\kappa = (\sqrt{2} C_{\text{AH}} G)^{-1},$$

the prefactor is equal to 1, so

$$e^{\lambda \mathcal{Q}(m)} - 1 + \sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \leq \sqrt{m} + \lambda \sqrt{m} e^{\lambda \mathcal{Q}(m)}.$$

Rearranging,

$$\sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \leq \sqrt{m} + 1 + (\lambda \sqrt{m} - 1) e^{\lambda \mathcal{Q}(m)}.$$

With

$$\lambda = \frac{1}{2\sqrt{T}} \quad \text{and} \quad m \leq T,$$

we have $\lambda\sqrt{m} \leq 1/2$, so the last term is nonpositive. Therefore,

$$\sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \leq \sqrt{m} + 1 \leq 2\sqrt{m}.$$

Since $\tilde{f}_t = \kappa f_t$,

$$\sum_{t=1}^m (f_t(p_t) - f_t(u^*)) \leq 2\kappa^{-1}\sqrt{m}.$$

Using

$$\kappa^{-1} = \sqrt{2} C_{\text{AH}} G = 2G\sqrt{2(4 + \ln K)},$$

we obtain

$$\sum_{t=1}^m (f_t(p_t) - f_t(u^*)) \leq 4G\sqrt{2(4 + \ln K)m}.$$

Finally, since

$$f_t(p_t) = \mathbb{E}_{I_t \sim p_t}[\omega_{t,I_t}], \quad f_t(u^*) = \langle \omega_t, u^* \rangle,$$

we have, for every $m \leq T$,

$$\sum_{t=1}^m \left(\mathbb{E}_{I_t \sim p_t}[\omega_{t,I_t}] - \langle \omega_t, u^* \rangle \right) \leq 4G\sqrt{2(4 + \ln K)m}.$$

Step 5: Cumulative excess-miscoverage bound.

By Assumption A and the fact that the ℓ_1 -diameter of Δ_K is 2,

$$\tilde{f}_t(p_t) - \tilde{f}_t(u^*) \geq -2\kappa G.$$

Thus,

$$\sum_{t=1}^m (\tilde{f}_t(p_t) - \tilde{f}_t(u^*)) \geq -2\kappa Gm.$$

Plugging this lower bound into the inequality from Step 4 before dropping the exponential term gives

$$e^{\lambda \mathcal{Q}(m)} (1 - \lambda\sqrt{m}) \leq 1 + \sqrt{m} + 2\kappa Gm.$$

Therefore,

$$\mathcal{Q}(m) \leq \frac{1}{\lambda} \ln \left(\frac{1 + \sqrt{m} + 2\kappa Gm}{1 - \lambda\sqrt{m}} \right).$$

For $\lambda = 1/(2\sqrt{T})$ and $m \leq T$,

$$1 - \lambda\sqrt{m} \geq \frac{1}{2},$$

so

$$\mathcal{Q}(m) \leq \frac{1}{\lambda} \ln \left(2(1 + \sqrt{m} + 2\kappa Gm) \right) \leq \frac{1}{\lambda} \ln \left(2 + (2 + 4\kappa G)m \right),$$

where we used $1 + \sqrt{m} \leq 1 + m$. Since

$$\kappa G = \frac{1}{\sqrt{2}C_{\text{AH}}}, \quad C_{\text{AH}} = 2\sqrt{4 + \ln K} \geq 4,$$

we have

$$4\kappa G = \frac{4}{\sqrt{2}C_{\text{AH}}} \leq \frac{\sqrt{2}}{2}.$$

Hence,

$$\mathcal{Q}(m) \leq \frac{1}{\lambda} \ln \left(2 + \left(2 + \frac{\sqrt{2}}{2} \right) m \right).$$

Finally, since

$$\mathcal{Q}(m) = \sum_{t=1}^m \kappa(g_t(p_t))_+,$$

we obtain

$$\sum_{t=1}^m (g_t(p_t))_+ = \kappa^{-1} \mathcal{Q}(m) \leq \frac{\kappa^{-1}}{\lambda} \ln\left(2 + \left(2 + \frac{\sqrt{2}}{2}\right)m\right).$$

Using

$$\frac{\kappa^{-1}}{\lambda} = 4G\sqrt{2(4 + \ln K)T},$$

we get, for every $m \leq T$,

$$\sum_{t=1}^m (g_t(p_t))_+ \leq 4G\sqrt{2(4 + \ln K)T} \ln\left(2 + \left(2 + \frac{\sqrt{2}}{2}\right)m\right).$$

Since

$$g_t(p_t) = \mathbb{E}_{I_t \sim p_t}[\text{err}_{t,I_t}] - \alpha,$$

this proves

$$\sum_{t=1}^m \left(\mathbb{E}_{I_t \sim p_t}[\text{err}_{t,I_t}] - \alpha \right)_+ \leq 4G\sqrt{2(4 + \ln K)T} \ln\left(2 + \left(2 + \frac{\sqrt{2}}{2}\right)m\right).$$

The proof is complete. $\square$

F Experiments

F.1 Methods

This section describes the implementation used in the synthetic and real-data experiments. All methods are implemented as online wrappers around a base predictor. In the regression experiments,

$$S_t = |Y_t - \hat{Y}_t|.$$

Online calibration windows. Each series is processed independently. Let

$$\mathcal{T}_{\text{test}} = (\tau_1, \dots, \tau_{T_{\text{test}}})$$

be the ordered test indices for a given series. At test position j , the calibration window contains the previous at most R test-time observations:

$$\mathcal{I}_j = \{\tau_{\max(1, j-R)}, \dots, \tau_{j-1}\}.$$

The first test point with an empty calibration window is skipped. This convention ensures that all calibration is online and uses no future outcomes.

CP. CP uses the finite-sample corrected rolling quantile of the previous calibration scores. If $r = |\mathcal{I}_j|$, then

$$k = \lceil (1 - \alpha)(r + 1) \rceil,$$

with k clipped to $\{1, \dots, r\}$, and

$$q_t^{\text{CP}} = \text{the } k\text{-th order statistic of } \{S_i : i \in \mathcal{I}_j\}.$$

The prediction set is $[\hat{Y}_t - q_t^{\text{CP}}, \hat{Y}_t + q_t^{\text{CP}}]$.

Localized weighted quantiles. LCP and OLC use weighted empirical quantiles. For a query covariate x , the calibration covariates in $\mathcal{I}_j$ are standardized coordinatewise using their empirical mean and standard deviation:

$$Z_i = \frac{X_i - \bar{X}_{\mathcal{I}_j}}{\hat{\sigma}_{\mathcal{I}_j}}, \quad Z_x = \frac{x - \bar{X}_{\mathcal{I}_j}}{\hat{\sigma}_{\mathcal{I}_j}},$$

where zero or numerically unstable standard deviations are replaced by 1. We then use the exponential localizer

$$H_h(x, X_i) = \exp\left(-\frac{\|Z_i - Z_x\|_2}{h}\right).$$

The weighted quantile is computed by sorting calibration scores and accumulating the corresponding sorted weights. If the total weight is numerically zero, the implementation falls back to uniform weights.

The base bandwidth is chosen by a Silverman-style rule,

$$h_0 = \left(\frac{4}{d+2}\right)^{1/(d+4)} R^{-1/(d+4)} \sqrt{d},$$

where d is the covariate dimension. The factor $(4/(d+2))^{1/(d+4)} R^{-1/(d+4)}$ is the classical multi-variate kernel bandwidth scaling [23, 22]. Because covariates are standardized within each calibration window before computing Euclidean distances, the additional $\sqrt{d}$ factor matches the typical scale of distances in d standardized dimensions. OLC-P-Hedge further reduces sensitivity to this heuristic by aggregating a grid of multiplicative bandwidths around h_0 .

LCP. LCP uses the localized empirical score distribution at the current covariate X_t ,

$$D_t^{(h_0)}(X_t) = \sum_{i \in \mathcal{I}_j} w_{t,i}^{(h_0)}(X_t) \delta_{S_i}, \quad w_{t,i}^{(h_0)}(X_t) = \frac{H_{h_0}(X_t, X_i)}{\sum_{r \in \mathcal{I}_j} H_{h_0}(X_t, X_r)}.$$

The fixed-level localized radius is

$$q_t^{\text{LCP}} = Q\left(1 - \alpha; D_t^{(h_0)}(X_t)\right),$$

and the prediction set is

$$[\hat{Y}_t - q_t^{\text{LCP}}, \hat{Y}_t + q_t^{\text{LCP}}].$$

ACI. ACI uses the rolling unweighted empirical score distribution, but replaces the fixed nominal level α by an adaptive level α_t . For a test stream of length T_{test} , the default step size is

$$\gamma = \frac{1}{2\sqrt{T_{\text{test}}}}.$$

At time t , ACI forms the rolling quantile at level $1 - \alpha_t$. After observing Y_t , it updates

$$\alpha_{t+1} = \Pi_{[0,1]}(\alpha_t + \gamma(\alpha - \text{err}_t)), \quad \text{err}_t = \mathbf{1}\{Y_t \notin \hat{C}_t\},$$

with initialization $\alpha_1 = \alpha$. Here we project the α_t back to $[0, 1]$ in the experiment to prevent infinite prediction sets for a better comparison.

DtACI. DtACI aggregates multiple ACI experts with different step sizes. Let

$$\gamma_0 = \frac{1}{2\sqrt{T_{\text{test}}}}, \quad \Gamma = \{0.25, 0.5, 0.75, 1, 1.25, 1.5\}\gamma_0.$$

Each expert r maintains its own adaptive level $\alpha_t^{(r)}$. The mixture level is

$$\alpha_t^{\text{DtACI}} = \sum_r p_{t,r} \alpha_t^{(r)}.$$

At time t , the method computes the empirical rank statistic

$$\beta_t = \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} \mathbf{1}\{S_i \geq S_t\}.$$

Expert weights are updated using exponential weights on the pinball loss

$$\ell(\beta_t, \alpha_t^{(r)}) = \alpha(\beta_t - \alpha_t^{(r)}) - \min\{0, \beta_t - \alpha_t^{(r)}\}.$$

Following [8], the implementation uses

$$I_{\text{size}} = 500, \quad \sigma_{\text{dt}} = \frac{1}{2I_{\text{size}}},$$

and

$$\eta_{\text{dt}} = \sqrt{\frac{3}{I_{\text{size}}}} \sqrt{\frac{\log(I_{\text{size}}|\Gamma|) + 2}{((1-\alpha)^2\alpha^3 + \alpha^2(1-\alpha)^3)/3}}.$$

The expert levels are then updated by their own projected ACI recursions.

SPCI. SPCI is implemented as a residual-forecasting conformal baseline. Let

$$\varepsilon_t = Y_t - \hat{Y}_t$$

denote the base-model residual on the online test stream. For each test time, we form a lag vector of the previous w_{lag} residuals and fit a quantile random forest to predict the next residual. The default parameters are

$$w_{\text{lag}} = 24, \quad T_{\text{train}} = R, \quad \text{refit_every} = 24, \quad \text{beta_grid_size} = 21.$$

The quantile random forest uses

$$\text{n_estimators} = 80, \quad \text{max_depth} = 10, \quad \text{min_samples_leaf} = 5, \quad \text{random_state} = 42, \quad \text{n_jobs} = -1.$$

For a grid

$$\beta \in \left\{0, \frac{\alpha}{20}, \frac{2\alpha}{20}, \dots, \alpha\right\},$$

SPCI predicts residual quantiles at levels β and $1 - \alpha + \beta$, and selects the value of β giving the narrowest prediction set. The final prediction set is

$$[\hat{Y}_t + \hat{q}_\beta, \hat{Y}_t + \hat{q}_{1-\alpha+\beta}].$$

OLCP. OLCP uses the localized family $C_t^{(h)}(X_t; \beta)$ from Section 3.1. In the experiments, $h = h_0$. At time t , OLCP forms

$$\hat{C}_t = C_t^{(h_0)}(X_t; \alpha_t),$$

where the quantile is the localized weighted quantile at level $1 - \alpha_t$. After observing Y_t , OLCP updates

$$\alpha_{t+1} = \Pi_{[0,1]}(\alpha_t + \gamma(\alpha - \text{err}_t)), \quad \text{err}_t = \mathbf{1}\{Y_t \notin \hat{C}_t\},$$

with

$$\gamma = \frac{1}{2\sqrt{T_{\text{test}}}}.$$

OLCP-Hedge. OLCP-Hedge aggregates OLCP experts over the bandwidth grid

$$h_i \in \{0.5, 0.75, 1, 1.25, 1.5\}h_0, \quad i = 1, \dots, 5.$$

Expert i maintains its own adaptive level $\alpha_{t,i}$ and outputs

$$C_{t,i}(X_t) = C_t^{(h_i)}(X_t; \alpha_{t,i}).$$

In regression, the set-size functional is interval width, so

$$\omega_{t,i} = \text{width}(C_{t,i}(X_t)),$$

and the expert miscoverage indicator is

$$e_{t,i} := \mathbf{1}\{Y_t \notin C_{t,i}(X_t)\}.$$

Each expert level is updated by

$$\alpha_{t+1,i} = \Pi_{[0,1]}(\alpha_{t,i} + \gamma(\alpha - e_{t,i})),$$

with

$$\gamma = \frac{1}{2\sqrt{T_{\text{test}}}}.$$

At round t , the expert cost is the prediction set size $\omega_{t,i}$, min-max normalized to $[0, 1]$ across experts, and all other parameters are chosen according to Section 3.2 with $G = 1$.

F.2 Running time on experiments

All experiments were run on a MacBook Pro. Neural network predictors were trained using Apple’s MPS backend when available. The conformal runtime table was measured on the same machine and reports only the online conformal calibration/evaluation step; it excludes data loading, base predictor training, forecast precomputation.

Table 3: Running time comparison. Entries report wall-clock time in seconds for the conformal calibration/evaluation step only. For simulation, we report the mean running time across 100 repetitions.

Method	Sim. A	Sim. B	Sim. C	ELEC2	ILINet	ETF volatility
CP	0.02	0.02	0.02	0.04	0.01	0.13
LCP	0.04	0.04	0.04	0.21	0.01	0.54
ACI	0.01	0.01	0.01	0.04	0.01	0.10
DtACI	0.02	0.02	0.02	0.11	0.01	0.20
SPCI	19.51	19.52	19.52	136.91	16.79	208.10
OLCP	0.05	0.05	0.05	0.26	0.01	0.41
OLCP-Hedge	0.09	0.09	0.09	0.60	0.04	0.75

F.3 Additional details and diagnostics for real-data experiments

This section provides implementation details and additional diagnostics for the real-data experiments in Section 4.3. We evaluate the same seven methods as in the synthetic experiments: CP, LCP, ACI, DtACI, SPCI, OLCP, and OLCP-Hedge. All methods are evaluated online with target miscoverage $\alpha = 0.1$ using rolling calibration windows. Reported sizes correspond to interval width for ELEC2 and ILINet; for ETF volatility, sizes are multiplied by 100 and reported in percentage points of absolute log return.

Implementation details. All experiments use a fixed point predictor followed by online conformal calibration. The point predictor is trained only on the training split and is not updated during conformal evaluation. The conformal methods use symmetric intervals of the form

$$[\hat{y}_t - q_t, \hat{y}_t + q_t],$$

with conformity score $S_t = |Y_t - \hat{y}_t|$. SPCI is implemented as a residual-autoregressive baseline using a sliding training window comparable to the conformal calibration window.

- **ELEC2.** ELEC2 contains electricity market prices, demands, and transfers from New South Wales and Victoria [11]. We use the normalized ELEC2 file and remove the initial constant-response segment, leaving 27,552 observations. The response is electricity transfer, and the covariates are `nswwprice`, `nswwdemand`, `vicprice`, and `vicedemand`. We keep the full half-hourly sequence. The first 70% of observations are used to train a fixed gradient-boosted regression tree predictor, implemented as `HistGradientBoostingRegressor` with maximum depth 6, learning rate 0.05, 400 boosting iterations, and random seed 42. Conformal methods are evaluated on the remaining 30% of the sequence with calibration window $R = 100$. For localized methods, the localization feature is the four-dimensional covariate vector above. For SPCI, we use residual lag length 24, training window R , refit frequency 24, and 21 candidate β -values.
- **ILINet.** ILINet is a weekly CDC influenza-like illness surveillance dataset [4, 12]. We use the weighted ILI component, which contains 1,305 weekly observations from October 1997 to October 2022. Missing values are filled by interpolation followed by forward/backward filling. The series is split chronologically into 70% training, 10% validation, and 20% testing, giving 913, 130, and 262 observations, respectively. The response is standardized using the training mean and standard deviation, and intervals are constructed on this standardized scale. The base predictor is a temporal convolutional network (TCN) [15] with input length 26, output length 1, batch size 32, at most 200 epochs, kernel size 5, 8 filters, dilation base 2, dropout 0.2, and Adam learning rate 10^{-3} . We use early stopping on validation loss with patience 3, minimum improvement 10^{-3} , gradient clipping at 0.1, and the best checkpoint. One-step forecasts are computed by historical forecasting with `forecast_horizon=1`, `stride=1`, and no retraining. For localized methods, X_t is the lag

window of the previous 26 standardized ILI values. The calibration window is $R = 52$. For SPCI, we use residual lag length 8, training window R , refit frequency 1, and 21 candidate β -values.

- **ETF volatility.** We forecast daily volatility proxies for five ETFs: SPY, QQQ, IWM, EEM, and TLT [25]. Daily closing prices are read from downloaded Stooq files, and the VIX index is obtained from FRED/CBOE [3]. The data are aligned to a business-day grid by forward-filling prices and VIX values. The response is the absolute daily log return

$$Y_t = |\log P_t - \log P_{t-1}|,$$

and the sample runs from January 2008 to March 2026. Each ETF series has 4,742 business-day observations after alignment. We use a chronological split with training through 2018, validation over 2019, and testing from 2020 onward, giving 2,868, 261, and 1,613 observations per asset. Each ETF volatility series is standardized using its own training mean and standard deviation; VIX is standardized using the VIX training split. The base predictor is a TCN trained jointly on the five standardized ETF volatility series, with input length 30, output length 1, batch size 256, at most 80 epochs, kernel size 5, 8 filters, dilation base 2, dropout 0.2, and Adam learning rate 10^{-3} . We include cyclic calendar encoders for day of week and month. Early stopping monitors validation loss with patience 5, minimum improvement 10^{-3} , gradient clipping at 0.1, and the best checkpoint. For localized methods, X_t consists of the previous 30 standardized volatility values concatenated with the lagged standardized VIX value, so the localization dimension is 31. The calibration window is $R = 200$. Since the TCN is trained on standardized responses, widths are converted back to the original absolute-log-return scale using the asset-specific training standard deviation and reported as $100 \times \text{width}$. For SPCI, we use residual lag length 30, training window R , refit frequency 5, and 21 candidate β -values.

Rolling diagnostics. Figures 2–4 show rolling coverage and rolling average size on the three real datasets. In each figure, the top panel shows rolling coverage and the bottom panel shows rolling average size. A desirable method stays close to the dashed 0.90 coverage line in the top panel while having a lower curve in the bottom panel.

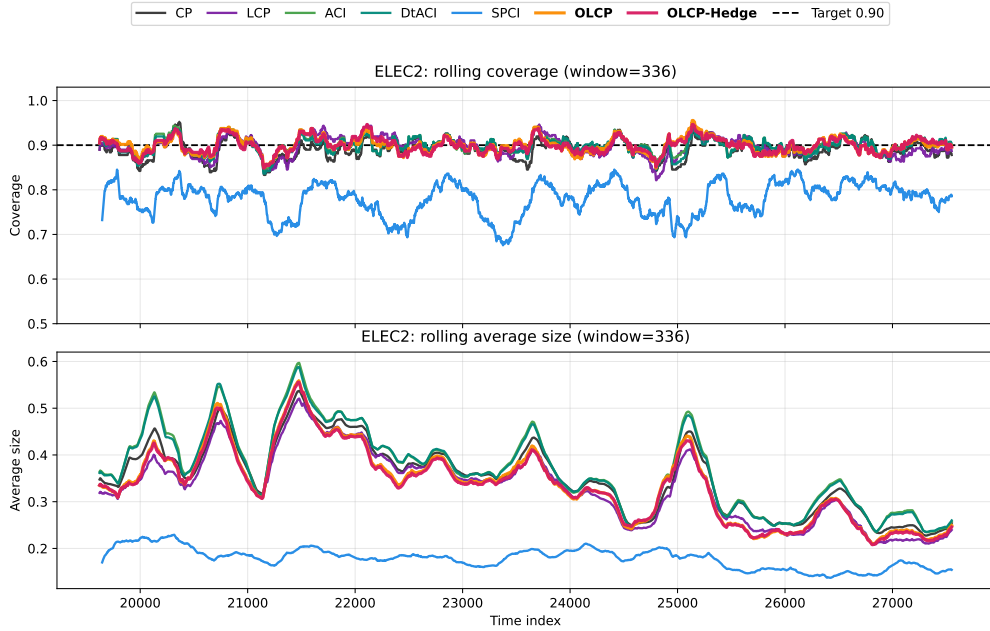


Figure 2: **ELEC2 rolling diagnostics.** Top: rolling coverage using a one-week window (48×7 half-hourly observations). Bottom: rolling average interval size using the same window. The dashed line marks target coverage 0.90.

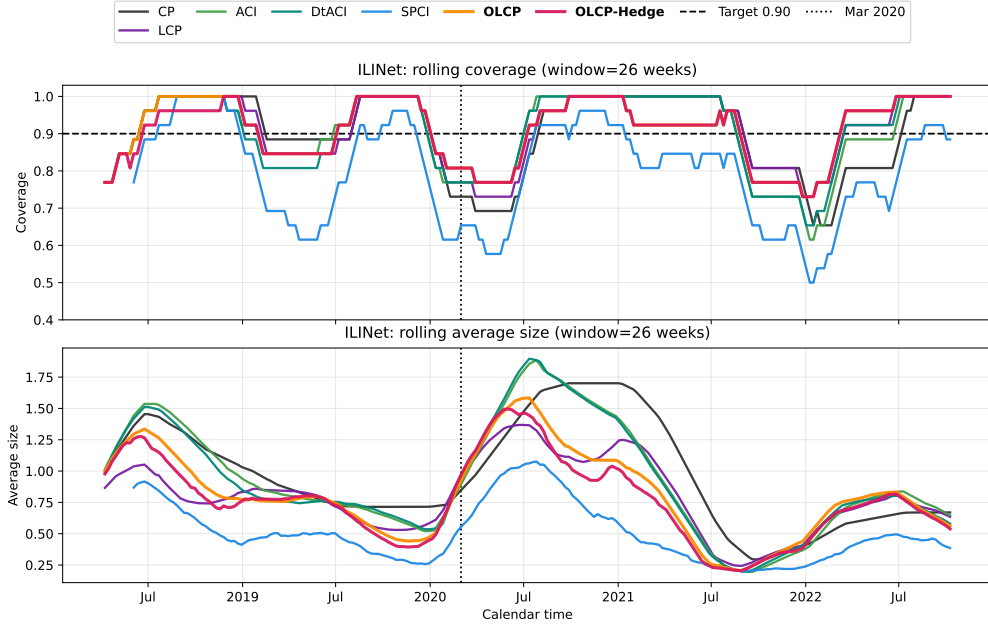


Figure 3: **ILINet rolling diagnostics.** Top: rolling coverage over weekly test observations. Bottom: rolling average interval size. The horizontal dashed line marks target coverage 0.90 while the vertical line marks the start of COVID.

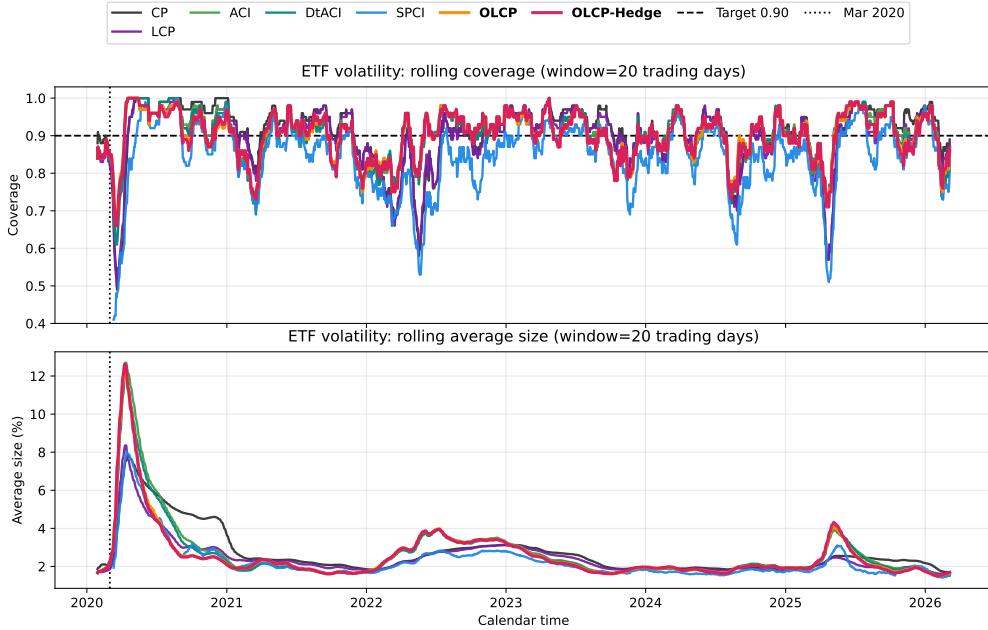


Figure 4: **ETF volatility rolling diagnostics.** Top: rolling coverage over daily test observations. Bottom: rolling average interval size, reported in percentage points of absolute log return. The dashed line marks target coverage 0.90.

Across the rolling diagnostics, SPCI is consistently much smaller but also persistently below the coverage target, indicating that its residual autoregression is not sufficiently calibrated in these nonstationary streams. ACI and DtACI tend to recover coverage by increasing sizes globally. OLCP and OLCP-Hedge are more efficient: their rolling sizes are generally below the global adaptive

baselines while their rolling coverage remains close to the target, especially outside the most extreme stress periods.

Volatility-regime diagnostics for ETF data. Table 4 stratifies ETF volatility performance by the current VIX level. This diagnostic evaluates whether methods adapt to market-volatility regimes rather than only achieving marginal coverage.

Table 4: **ETF volatility diagnostics by VIX regime.** Low- and high-VIX regimes are defined as the bottom and top quartiles of VIX over the online test period: $VIX \leq 15.87$ and $VIX \geq 23.84$, respectively. Sizes are reported in percentage points of absolute log return. N is the number of evaluated ETF-day prediction points in each regime; SPCI has fewer low-VIX points because of its additional residual-lag warm-up.

Method	Low VIX			High VIX		
	Coverage	Size	N	Coverage	Size	N
CP	0.925	2.044	2015	0.854	3.835	2015
LCP	0.920	1.900	2015	0.846	3.356	2015
ACI	0.913	1.851	2015	0.890	4.349	2015
DtACI	0.915	1.854	2015	0.889	4.184	2015
SPCI	0.862	1.693	1895	0.792	3.272	2015
OLCP	0.912	1.833	2015	0.890	4.004	2015
OLCP-Hedge	0.913	1.824	2015	0.888	4.007	2015

The VIX-stratified table shows that high-volatility periods are substantially harder: all methods have lower conditional coverage when VIX is high. Global adaptive methods improve high-VIX coverage by inflating sizes, whereas OLCP and OLCP-Hedge achieve comparable or slightly better high-VIX coverage with smaller sizes than ACI and DtACI. SPCI remains the narrowest method but undercovers severely in both regimes. This supports the main empirical conclusion: localization improves efficiency, online calibration helps maintain validity, and their combination gives the best overall coverage–size tradeoff.